Discrete Structures

Topic 2: Sets

2 · What is a Set?

- **Set: a collection of distinct, well-defined objects.**

  Well-defined means we can always decide whether an object is in the set or not.

- **Written with braces: A = {1, 2, 3}. The objects are called elements.**
- **Order does not matter: {1, 2, 3} = {3, 2, 1}.**

  A set only records WHAT is inside, not where.

- **Repeats are ignored: {1, 1, 2} = {1, 2}.**

  An element is either in a set or not; being in twice means nothing.

2 · Membership, Notation & Size

- **$x \in A$ means "x belongs to A". $x \notin A$ means "x does not belong".**

  A = {2, 4, 6}: $4 \in A$ is true, $5 \notin A$ is true.

- **Roster form: list the elements, e.g. {2, 4, 6}.**
- **Set-builder form: {x | x is even and 0 < x < 8}.**

  Read it as "all x such that ...". Useful when the list is long or infinite.

- **|A| is the number of elements (cardinality). |{2, 4, 6}| = 3.**

2 · Standard Number Sets

- **$\mathbb{N} = \{0, 1, 2, 3, \ldots\}$ natural numbers**

  Some books start $\mathbb{N}$ at 1. Always check your textbook's convention.

- **$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$ integers**
- **$\mathbb{Q}$ = all fractions a/b with integers a, b and $b \neq 0$ (rational)**
- **$\mathbb{R}$ = all real numbers (every point on the number line)**
- **Relation: $\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$**

  Every natural number is an integer, every integer is a fraction ($5 = 5/1$), and so on.

2 · The Empty Set and Universal Set

- **Empty set: ∅ = { } has NO elements, so |∅| = 0.**

  Example: {x | x is an integer and $x^2 = -1$} = ∅ because no integer squares to −1.

- **{∅} is NOT empty. It has ONE element: the empty set.**

  Think of boxes: an empty box vs a box that contains an empty box.

- **Universal set U: the set of everything we are talking about in the problem.**

  Example: if we study digits, U = {0, 1, ..., 9}.

2 · Equal Sets and Subsets

- **$A = B$ if they have exactly the same elements.**
- **$A \subseteq B$ (subset): every element of A is also in B.**

  $\{1, 2\} \subseteq \{1, 2, 3\}$ because 1 and 2 are both in $\{1, 2, 3\}$.
- **$A \subset B$ (proper subset): $A \subseteq B$ and $A \neq B$.**
- **$\emptyset \subseteq A$ for every set A, and $A \subseteq A$.**

  Why? To fail, $\emptyset$ would need an element that is not in A. But $\emptyset$ has no elements, so it cannot fail.

2 · Example: ∈ vs ⊆

Let A = {1, 2, 3}.

1 ∈ A TRUE

1 is an element of A.

{1} ⊆ A TRUE

The set {1} has the element 1, which is in A.

{1} ∈ A FALSE

The elements of A are numbers, not sets. {1} is not one of them.

4 ∉ A TRUE, {1, 4} ⊆ A FALSE (4 is missing)

2 · Power Set

- **Power set P(A): the set of ALL subsets of A.**
  It always contains ∅ and A itself.
- **P({a, b}) = { ∅, {a}, {b}, {a, b} }**
- **If |A| = n, then |P(A)| = 2^n.**
  Why? Building a subset, each of the n elements has 2 choices: in or out. So 2 × 2 × ... × 2 = 2^n.
- **Check: {a, b} has n = 2, and 2^2 = 4 subsets ✓**

2 · Example: Power Set of {1, 2, 3}

n = 3, so we expect $2^3 = 8$ subsets.

Size 0: ∅

Size 1: {1}, {2}, {3}

Size 2: {1, 2}, {1, 3}, {2, 3}

Size 3: {1, 2, 3}

Count: 1 + 3 + 3 + 1 = 8 ✓

Listing by size makes sure no subset is missed.

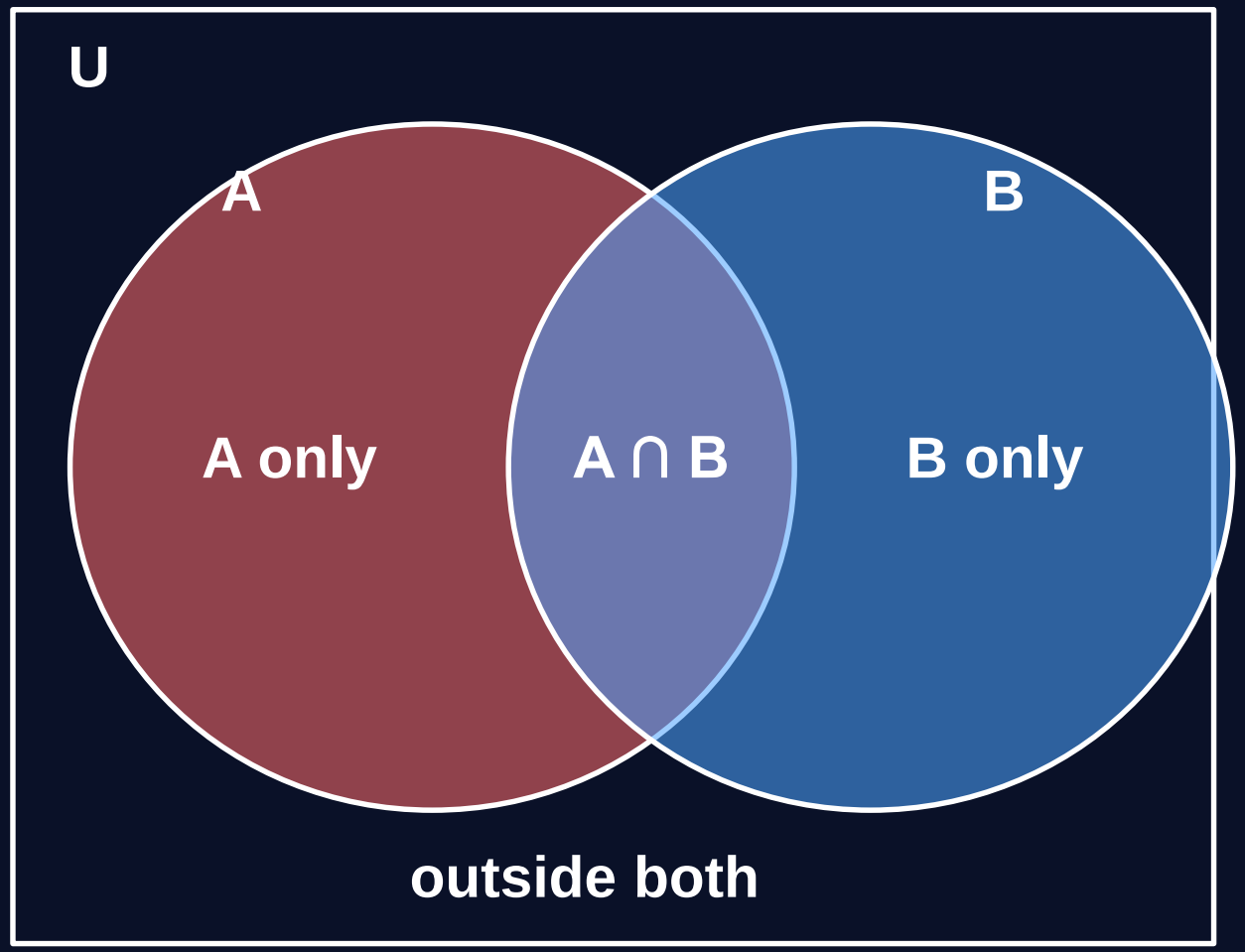
2 · Union and Intersection

- **Union $A \cup B = \{x \mid x \in A \text{ OR } x \in B\}$**
  Everything that is in at least one of the sets.
- **Intersection $A \cap B = \{x \mid x \in A \text{ AND } x \in B\}$**
  Only what is in both sets.
- **$A = \{1, 2, 3\}$, $B = \{3, 4\}$**
- **$A \cup B = \{1, 2, 3, 4\}$ $A \cap B = \{3\}$**
- **If $A \cap B = \emptyset$, the sets are called disjoint (nothing in common).**

2 · Difference and Complement

- **Difference $A - B = \{x \mid x \in A \text{ AND } x \notin B\}$**
  Take A and remove everything that also lies in B.
- **Complement $A' = U - A$**
  Everything in the universal set that is NOT in A.
- **$U = \{1, \ldots, 10\}$, $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5\}$**
- **$A - B = \{1, 2\}$ $B - A = \{5\}$**
- **$A' = \{5, 6, 7, 8, 9, 10\}$**
  Note $A - B \neq B - A$. Order matters for difference.

2 · Venn Diagram of A and B



- Rectangle = universal set U.
- Left circle = A, right circle = B.
- Overlap = A ∩ B.
- All coloured area = A ∪ B.
- Left part only = A − B.
- Outside both circles = (A ∪ B)′.

2 · Example: All Operations Together

$U = \{1, \ldots, 8\}$, $A = \{1, 2, 3, 4, 5\}$, $B = \{4, 5, 6, 7\}$

$A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$

$A \cap B = \{4, 5\}$

$A - B = \{1, 2, 3\}$ $\quad$ $B - A = \{6, 7\}$

$A' = \{6, 7, 8\}$

$(A \cup B)' = \{8\}$

Always check: $(A \cup B)'$ contains what is in U but in neither A nor B. Only 8 qualifies.

2 · Set Laws (Identities)

- **Commutative: $A \cup B = B \cup A$, $A \cap B = B \cap A$**
- **Distributive: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$**
- **Complement: $A \cup A' = U$, $A \cap A' = \emptyset$**
- **De Morgan: $(A \cup B)' = A' \cap B'$**
- **De Morgan: $(A \cap B)' = A' \cup B'$**

De Morgan in words: NOT (in A or B) = not in A AND not in B. These match the logic laws you will meet later.

2 · Example: Check De Morgan with Numbers

$U = \{1, \ldots, 6\}$, $A = \{1, 2, 3\}$, $B = \{3, 4\}$

$A \cup B = \{1, 2, 3, 4\}$ so $(A \cup B)' = \{5, 6\}$

$A' = \{4, 5, 6\}$, $B' = \{1, 2, 5, 6\}$

$A' \cap B' = \{5, 6\}$

Both sides are {5, 6}, so the law holds here ✓

A numerical check builds confidence, but it is not a proof.

The next slides show how to prove it.

2 · How to Prove Statements About Sets

- **To prove $A \subseteq B$:**
  Take an arbitrary element $x \in A$ and show that $x \in B$.
- **To prove $A = B$:**
  Prove $A \subseteq B$ and also $B \subseteq A$.
- **Use the definitions: $x \in A \cap B$ means $x \in A$ AND $x \in B$; $x \in A \cup B$ means $x \in A$ OR $x \in B$.**
  "Arbitrary" means x is not a special element, so the argument works for every element.

2 · Example: Two Small Proofs

Prove $A \cap B \subseteq A$.

Let $x \in A \cap B$. Then $x \in A$ and $x \in B$ (definition of $\cap$).

So $x \in A$ [QED]

Prove $A \subseteq A \cup B$.

Let $x \in A$. Then "$x \in A$ or $x \in B$" is true because its first part is true.

So $x \in A \cup B$ [QED]

An OR statement is true if at least one part is true.

2 · Cartesian Product

- **$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$ (ordered pairs)**
  In an ordered pair the order matters: $(1, 2) \neq (2, 1)$.
- **$A = \{1, 2\}$, $B = \{x, y\}$**
- **$A \times B = \{(1,x), (1,y), (2,x), (2,y)\}$**
- **$|A \times B| = |A| \cdot |B|$**
  Why? Each of the $|A|$ first parts pairs with each of the $|B|$ second parts. Here $2 \times 2 = 4$ ✓
- **Used later for relations and databases (rows of a table).**

2 · Counting Elements of a Union

- **$|A \cup B| = |A| + |B| - |A \cap B|$**

  Elements in both sets are counted twice when adding |A| + |B|, so we subtract them once.

- **Example: 20 students play cricket, 15 play football, 8 play both.**
- **Students who play at least one game: 20 + 15 − 8 = 27**

  Check: cricket only = 12, football only = 7, both = 8. And 12 + 7 + 8 = 27 ✓

2 · Summary

- A set is a collection of distinct elements; order and repeats do not matter.
- $\in$ is for elements; $\subseteq$ is for subsets.
- $|P(A)| = 2^n$, $|A \times B| = |A| \cdot |B|$.
- $\cup$ = OR, $\cap$ = AND, $A - B$ = in A not in B, A' = not in A.
- $|A \cup B| = |A| + |B| - |A \cap B|$.
- To prove $A = B$, show $A \subseteq B$ and $B \subseteq A$.

2 · Practice Questions

Q1. $A = \{1, 3, 5, 7\}$, $B = \{3, 4, 5\}$, $U = \{1, \ldots, 8\}$. Find $A \cup B$, $A \cap B$, $A - B$, A'.

Q2. How many subsets does {a, b, c, d} have? Why?

Q3. True or false: $\emptyset \subseteq \{1\}$? $\emptyset \in \{1\}$?

Q4. In a class, 30 like maths, 25 like physics, 10 like both. How many like at least one?

Q5. Prove $A \cap B \subseteq B$.

2 · Solutions

Q1. $A \cup B = \{1,3,4,5,7\}$, $A \cap B = \{3,5\}$, $A - B = \{1,7\}$, $A' = \{2,4,6,8\}$

Q2. $2^4 = 16$, since each of the 4 elements is either in or out.

Q3. $\emptyset \subseteq \{1\}$ is TRUE ($\emptyset$ is a subset of every set). $\emptyset \in \{1\}$ is FALSE (the only element of $\{1\}$ is 1).

Q4. $30 + 25 - 10 = 45$

Q5. Let $x \in A \cap B$. Then $x \in A$ and $x \in B$, so $x \in B$ [QED]